

DAT BUI

Lincoln, NE | (267) 965-7441 | quangdatbui10112004@gmail.com | linkedin.com/in/datbui1011

EDUCATION

University of Nebraska–Lincoln

B.S. Computer Science — GPA: 3.924 / 4.0

Lincoln, NE

Expected May 2027

WORK EXPERIENCE

Beavercreek Marketing

Senior Software Developer Intern

Lincoln, NE

Feb 2026 – Present

- Built the distributed video rendering platform architecture using **Cloudflare Workers** and **JavaScript** — cutting rendering time by **40%**, enabling dead-letter recovery, and scaling to 5,000+ msg/sec throughput.
- Owned a multi-process Mac Mini render farm orchestrating **After Effects** and **FFmpeg** — eliminating race conditions across automated branded video pipelines for 3,000+ financial institutions.
- Built an internal admin portal and B2B client portal with **React**, **Cloudflare Workers** AI caption integration, and real-time monitoring.
- Mentored an incoming intern through codebase onboarding and contribution workflows.

UNL – School of Computing

Undergraduate Researcher

Lincoln, NE

Oct 2025 – Present

- Conducting research on Explainable AI using Topological Laplacians with **PyTorch**, **giotto-tda**, **NumPy**, **scikit-learn**, and **Jupyter Notebook**.
- Built and evaluated deep learning pipelines on 3+ benchmark datasets, integrating persistent Laplacians and persistence diagram vectorization to analyze latent feature representations.
- Improving the transparency and interpretability of deep learning models by applying topological data analysis techniques to hidden-layer representations.

KZValve, LLC

IT Systems Support Intern

Greenwood, NE

May 2025 – Feb 2026

- Cut manual work by **30%** and delivered an estimated **\$5,000/year** in cost savings by designing AI-powered automation across Power Automate, Power Apps, Power BI, and MS SQL Server.
- Automated weekly processing of **10,000+ records**, compressing insight turnaround by **35%** and enabling same-day decision-making instead of end-of-week reporting.
- Reduced reporting time by **50%** by building real-time alert workflows for data changes and errors, giving ops teams continuous visibility without manual monitoring.

UNL – School of Computing

Learning Assistant

Lincoln, NE

Jan 2025 – Present

- Led weekly lab sections for **20 students**, reinforcing core **OOP concepts** through hands-on Java exercises that turned abstract principles into working, debuggable code.

PROJECTS

L.U.C.A. Coaching | React, Python, Supabase

Sep 2025 – May 2026

- Architected a PWA to bridge the knowledge gap for newly graduated nurses, delivering personalized coaching at the point of clinical decision-making.
- Built an ML inference pipeline analyzing patient vital signs in real time, triggering contextual coaching prompts that reinforce clinical judgment under pressure.

DB Cinema | MongoDB, Express, React, Node.js, Vite

Sep 2025 – Dec 2025

- Built a full-stack cinema ticketing platform deployed on Vercel + MongoDB Atlas — achieving **180ms** API response times, **99.9% uptime**, and stability under **150+ concurrent users**.
- Automated the booking pipeline with Inngest + Resend API, cutting email delivery time by **90%** and issuing **200+ QR-coded** tickets for gate validation.

TECHNICAL SKILLS

Languages	Python, Java, C, C++, C#, JavaScript, TypeScript, Swift, SQL
Web & Frameworks	Bootstrap, React, Node.js, FastAPI, ASP.NET, REST APIs.
Databases	PostgreSQL, MySQL, MongoDB, MS SQL Server, Supabase
AI / ML	PyTorch, scikit-learn, NumPy, giotto-tda
Cloud & DevOps	Cloudflare, Vercel, AWS, Google Cloud
AI tool	Claude Code, GitHub Copilot